

# Vasudev Malyan

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## EDUCATION

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| JUL 2019 –<br>AUG 2026 | <b>MSc - PhD in Environmental Science and Engineering</b><br><i>Indian Institute of Technology Bombay, Mumbai, India, <u>MSc Grade: 9.68/10</u></i> <ul style="list-style-type: none"><li>Received the <b>Prime Minister's Research Fellowship</b> (2023–25), awarded to a small national cohort of doctoral students across Indian institutions</li><li>Received the <b>Late Shri Girish Vishnuprasad Desai Award for Academic Excellence</b> in 2023</li></ul> |
| JUL 2016 –<br>MAY 2019 | <b>BSc in Physical Science</b><br><i>University of Delhi, Delhi, India, <u>BSc Grade: 8.21/10</u> (Division: First)</i>                                                                                                                                                                                                                                                                                                                                          |

## PROFESSIONAL EXPERIENCE

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| JUL 2026 –<br>PRESENT  | <b>Technical Expert – Air Quality Sensor Technologies, Ubiquedge Technologies Pvt. Ltd., India</b> <ul style="list-style-type: none"><li>Heading <b>R&amp;D</b>, defining the technical roadmap and research priorities for <b>sensor calibration</b>, <b>source apportionment</b>, and <b>new product development</b> within the air quality sensing portfolio</li><li>Leading development of <b>network-scale calibration models</b> and <b>source characterization</b> for an ongoing low-cost sensor deployment across <b>150+ construction sites</b> in Maharashtra, India</li><li>Developing a <b>source-informed calibration engine</b> that classifies site typology from sensor-native features and applies source-stratified calibration</li></ul> |
| JAN 2026 –<br>MAY 2026 | <b>Research Associate, Indian Institute of Technology Bombay, Mumbai, India</b> <ul style="list-style-type: none"><li>Built a <b>physics-based calibration framework</b> for low-cost sensors that reduces dependence on co-located reference monitors, tested from chamber to field deployment</li><li>Developed a <b>probabilistic Mie-scattering model</b> coupling <math>\kappa</math>-Köhler hygroscopic growth with sensor-specific optical response, enabling calibration without co-located reference monitors</li><li>Validated the framework against <b>chamber experiments</b> and <b>field deployments</b> spanning multiple aerosol types and humidity regimes</li></ul>                                                                        |

## RESEARCH EXPERIENCE

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| JUL 2021 –<br>MAY 2026 | <b>PhD Project: Calibration of low-cost particulate matter sensors using physics-based and machine learning frameworks – from laboratory evaluation to field applications</b><br><i>PI: Prof. Manoranjan Sahu, Indian Institute of Technology Bombay, India</i> <ul style="list-style-type: none"><li>Completed PhD with <b>5 first-author</b> and <b>7 co-authored publications</b></li><li>Performed <b>controlled chamber experiments</b> across aerosol types, size distributions, and humidity regimes to isolate the physical mechanisms driving sensor error, published as a two-part series in <i>ACS ES&amp;T Air</i> assessing both number- and mass-concentration response</li><li>Developed a <b>physical-optical model</b> coupling Mie scattering with <math>\kappa</math>-Köhler hygroscopic growth theory, an inexpensive route to calibrating sensors without a co-located regulatory monitor</li><li>Established that <b>aerosol source</b> and <b>size distribution</b> govern whether a calibration model transfers between sites using data from <b>field campaigns</b> across NCT-Delhi and Mumbai</li><li>Experience with <b>aerosol measurement</b> (SMPS-CPC, BAM-1020, GRIMM 11-D and OPS 1.109, TSI DustTrak, PurpleAir PA-II and Alphasense OPC-N2/N3), and <b>generation</b> (PALAS Aerosol Dust Generator, TSI Jet Nebulizer) systems</li><li>Extended the work into policy, co-authoring a <b>T20 brief</b> on sensor-satellite synergies for addressing the issues related to ambient air quality monitoring in G20 countries</li><li><b>Oral presenter</b> at the National Environment Conference (2024), Indian Aerosol Science and Technology Association Conference (2023), and Air Sensors International Conference (2022)</li><li>Invited <b>peer reviewer</b> for <b>Atmospheric Pollution Research</b></li></ul> |
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| FEB 2022 –<br>APR 2023 | <p><b>IoT-based low-cost sensor network for air quality monitoring in IIT Bombay campus</b></p> <p><i>PI: Prof. Manoranjan Sahu, Indian Institute of Technology Bombay, India</i></p> <ul style="list-style-type: none"> <li>Designed and implemented the <b>IoT-based LCS network architecture</b> across the IIT Bombay campus, developed <b>calibration algorithms</b>, and integrated them into the project software</li> <li>Managed collaboration between the software development team and institute stakeholders</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| DEC 2021 –<br>APR 2022 | <p><b>Pilot study on reducing the particulate air pollution in urban areas using large-scale air cleaning system (“Smog Tower”)</b></p> <p><i>PI: Prof. Manoranjan Sahu, Indian Institute of Technology Bombay, India</i></p> <ul style="list-style-type: none"> <li>Built an <b>end-to-end analysis pipeline</b> integrating reference monitors and low-cost sensors to evaluate impact of smog-tower on hyperlocal air quality at two sites in NCT-Delhi</li> <li>Designed automated worksheets to process and visualize intrinsic and extrinsic filter differential pressure and track performance targets of the smog towers</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| JUL 2019 –<br>JUL 2021 | <p><b>MSc Project: Techno-economic study of air pollution control technologies for capturing particulate matter</b></p> <p><i>PI: Prof. Manoranjan Sahu, Indian Institute of Technology Bombay, India</i></p> <ul style="list-style-type: none"> <li>Evaluated <b>industrial particulate control technologies</b> including electrostatic precipitators, baghouse filters, cyclones, and scrubbers against capture efficiency, capital cost, and operating cost to produce selection guidance for industrial abatement</li> <li>Built the analysis around <b>particle size-resolved collection efficiency</b> rather than bulk removal rates, so that technology choice could be matched to the emitting process</li> <li>Extended the review to <b>indoor PM control</b>, comparing filtration and air-cleaning technologies for occupational settings, published in <i>Aerosol Science and Engineering</i></li> <li>Master’s seminar report published as a <b>book chapter with Springer Nature</b> on technologies for controlling particulate emissions from industry</li> <li>Achieved <b>highest CPI</b> in the MSc batch (9.68/10), recognized with the Late Shri Girish Vishnuprasad Desai Award for Academic Excellence</li> </ul> |

## PUBLICATIONS

13 total, 6 first-author, citations = 154, h-index = 7, i10-index = 5

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| PEER-<br>REVIEWED | <p><b>Malyan, V., &amp; Sahu, M.,</b> “Development and Assessment of a Physical-optical Calibration Model for Low-cost Sensors”, <i>Manuscript under review</i> (2026).</p> <p><b>Malyan, V., Sahu, M., &amp; Nawale, P.,</b> “Design and Evaluation of a Calibration Chamber for Low-Cost PM Sensors – Part 2: Mass Concentration-Based Assessment”, <i>ACS ES&amp;T Air</i>, 3(6), 1677-1690 (2026). (<a href="#">Link</a>)</p> <p><b>Malyan, V., Nawale, P., &amp; Sahu, M.,</b> “Design and Evaluation of a Calibration Chamber for Low-Cost PM Sensors – Part 1: Number Concentration-Based Assessment”, <i>ACS ES&amp;T Air</i>, 3(1), 129-141 (2025). (<a href="#">Link</a>)</p> <p>Zaid, M., Nawale, P., Kumar, V., <b>Malyan, V., &amp; Sahu, M.,</b> “Investigating IoT-Based Low-cost Sensor Network for Real-Time Hyper-Local Air Quality Monitoring and Exposure Assessment”, <i>Atmospheric Pollution Research</i>, 17(2), 102749 (2025). (<a href="#">Link</a>)</p> <p><b>Malyan, V., Kumar, V., Moni, M., Sahu, M., Prakash, J., Choudhary, S., Raliya, R., Chadha, T. S., Fang, J., &amp; Biswas, P.,</b> “Assessing the Spatial Transferability of Calibration Models across a Low-cost Sensors Network”, <i>Journal of Aerosol Science</i>, 181, 106437 (2024). (<a href="#">Link</a>)</p> <p>Kumar, V., <b>Malyan, V., Sahu, M., &amp; Biswal, B.,</b> “Aerosol Sources Characterization and Apportionment from Low-Cost Particle Sensors in an Urban Environment”, <i>Atmospheric Environment: X</i>, 22, 100271 (2024). (<a href="#">Link</a>)</p> <p><b>Malyan, V., Kumar, V., Sahu, M., Prakash, J., Choudhary, S., Raliya, R., Chadha, T. S., Fang, J., &amp; Biswas, P.,</b> “Calibrating Low-cost Sensors Using MERRA-2 Reconstructed PM2.5 Mass Concentration as a Proxy”, <i>Atmospheric Pollution Research</i>, 15(3), 102027 (2023). (<a href="#">Link</a>)</p> |
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Kumar, V., **Malyan, V.**, Sahu, M., Biswal, B., Pawar, M., & Dev, I., “Spatiotemporal Analysis of Fine Particulate Matter for India (1980–2021) from MERRA-2 Using Ensemble Machine Learning”, *Atmospheric Pollution Research*, 14(8), 101834 (2023). ([Link](#))

Kumar, V., **Malyan, V.**, Sahu, M., & Biswal, B., “Machine Learning Classification Model to Label Sources Derived from Factor Analysis Receptor Models for Source Apportionment”, *Aerosol and Air Quality Research*, 23(7), 220386 (2023). ([Link](#))

Dharaiya, V. R., **Malyan, V.**, Kumar, V., Sahu, M., Venkataraman, C., Biswas, P., *et al.*, “Evaluating the Performance of Low-cost PM Sensors over Multiple COALESCE Network Sites”, *Aerosol and Air Quality Research*, 23(5), 220390 (2023). ([Link](#))

Kumar, A., **Malyan, V.**, & Sahu, M., “Air Pollution Control Technologies for Indoor Particulate Matter Pollution: A Review”, *Aerosol Science and Engineering*, 7(2), 261–282 (2023). ([Link](#))

**Malyan, V.**, Kumar, V., & Sahu, M., “Significance of Sources and Size Distribution on Calibration of Low-cost Particle Sensors: Evidence from a Field Sampling Campaign”, *Journal of Aerosol Science*, 168, 106114 (2023). ([Link](#))

Kumar, V., **Malyan, V.**, & Sahu, M., “Significance of Meteorological Feature Selection and Seasonal Variation on Performance and Calibration of a Low-Cost Particle Sensor”, *Atmosphere*, 13(4), 587 (2022). ([Link](#))

#### CHAPTER & POLICY BRIEF

**Malyan, V.**, Kumar, V., Sahu, M., Mayya, Y. S., Biswas, P., & Venkataraman, C., “Improving Air Quality Monitoring in G20 Countries through Low-Cost Sensor–Satellite Synergies”, *T20 Policy Brief* (2023).

Sahu, M., **Malyan, V.**, & Mayya, Y. S., “Technologies for Controlling Particulate Matter Emissions from Industries”, in *Pollution Control Technologies* (Energy, Environment, and Sustainability), Springer, Singapore (2021). ([Link](#))

#### PEER REVIEW

Invited **peer reviewer** for **Atmospheric Pollution Research**

### CONFERENCE PRESENTATIONS

#### ORAL

“Calibrating Low-cost Sensors Using MERRA-2 Reconstructed PM<sub>2.5</sub> Mass Concentration as a Proxy”, National Environment Conference (NEC’24), Mumbai, India (2024)

“Optimizing Reference Co-location Sites for Improved Transferability of Calibration Models within a Low-cost Sensor Network in NCT-Delhi”, IASTA Conference, Navi Mumbai, India (2023)

“Fundamental Issues with Low-Cost Particle Sensors: Evidence from a Field Sampling Campaign”, 2nd Air Sensors International Conference (ASIC), Bangalore, India (2022) – **Student Travel Award**, UC Davis AQRC

“Significance of Meteorological Feature Selection and Seasonal Variation on Performance and Calibration of a Low-Cost Particle Sensor”, 12th Asian Aerosol Conference, Virtual (2022)

#### POSTER

“Low-cost Particulate Matter Sensor Evaluation in a Calibration Chamber: Number and Mass Concentration Assessment”, 14th Asian Aerosol Conference (AAC’25), Mumbai, India (2025) – lightning talk + poster

“Assessing the Spatial Transferability of Calibration Models across a Low-cost Sensors Network”, Air Sensors International Community Conference, Accra, Ghana (2023)

“Distinct Seasonal Effects of Hygroscopicity on the Performance of Low-cost Particle Sensors”, 5th India Clean Air Summit (ICAS), Bangalore, India (2023) – **Student Travel Award**, CSTEP

“Significance of Sources and Size Distribution on Calibration of Low-cost Particle Sensors: Evidence from a Field Sampling Campaign”, EGU General Assembly, Vienna, Austria (2023)

“Performance Comparison of Different Particulate Matter Control Technologies in Indoor Environments”, Indoor Air 2022, Kuopio, Finland (2022)

“Comparative Analysis of Different Indoor Particle Control Technologies”, 6th Indian International Conference on Air Quality Management, Virtual (2021)

## TEACHING EXPERIENCE

|                     |                                                                                                                                                                                                                                                                                                                                                                      |
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| 2021 – 2025         | <b>Teaching Assistant</b> , Indian Institute of Technology Bombay<br>Environmental Studies (ES200, Spring 2021), Data Analysis and Interpretation (ES209, 2021/22/24), Air Pollution Control Technologies (ES672, 2022/23/24), Environmental Monitoring Laboratory (ES651, Fall 2023), and Computational Laboratory for Environmental Engineers (ES319, Spring 2025) |
| MAY 2025 – JUL 2025 | <b>PMRF Teaching Assistantship</b><br><b>Guest Lecturer</b> , Delhi Institute of Rural Development, Delhi, India   28 Lectures<br>Delivered lectures on Environmental Studies: Science and Engineering to undergrad students                                                                                                                                         |
| AUG 2024 – NOV 2024 | <b>Guest Lecturer</b> , AKS University, Madhya Pradesh, India   32 Lectures<br>Delivered lectures on Environmental Chemistry to postgrad students across a full semester, covering the fundamentals of atmospheric chemistry                                                                                                                                         |
| JAN 2024 – APR 2024 | <b>Guest Lecturer</b> , Sandesh College of Arts, Commerce and Science, Mumbai, India   36 Lectures<br>Delivered lectures on Fundamentals of Organic Chemistry - Reaction Mechanisms to undergrad students across a full semester                                                                                                                                     |
| JAN 2023 – APR 2023 | <b>Guest Lecturer</b> , Sandesh College of Arts, Commerce and Science, Mumbai, India   25 Lectures<br>Delivered lectures on Calculus II to undergrad students across a full semester                                                                                                                                                                                 |
| 2022                | <b>Guest Lecture</b> , Indian Institute of Technology Bombay<br>Delivered a 1 hour lecture on air pollution control technologies (ESP and baghouse filters) in ES317 Fundamentals of Air Pollution Science and Engineering to BTech + MTech Dual Degree students                                                                                                     |

## AWARDS & FELLOWSHIPS

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| 2023 – 25 | <b>Prime Minister's Research Fellowship</b> , Indian Institute of Technology Bombay, India           |
| 2023      | <b>Late Shri Girish Vishnuprasad Desai Award for Academic Excellence</b> , IIT Bombay, India         |
| 2023      | <b>Student Travel Award</b> , Center for Study of Science, Technology and Policy, India              |
| 2022      | <b>ASIC-India Travel Support</b> , Air Quality Research Center, University of California, Davis, USA |
| 2019 – 22 | <b>Scholarship for MSc-PhD Dual Degree Programme</b> , Indian Institute of Technology Bombay, India  |
| 2016      | <b>Fourth Place</b> , Intel International Science and Engineering Fair, Phoenix, USA                 |
| 2015      | <b>First Place</b> , Initiative for Research and Innovation in STEM (IRIS) National Fair, India      |
| 2014      | <b>First Place</b> , Initiative for Research and Innovation in STEM (IRIS) National Fair, India      |
| 2014      | <b>Special Award</b> , Gujarat Council on Science and Technology, India                              |
| 2013      | <b>First Place</b> , Initiative for Research and Innovation in STEM (IRIS) National Fair, India      |

## PERSONAL PROJECTS

|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| SEP 2019 – APR 2020 | <b>Installation of aerators on taps</b> , Sustainability Cell, IIT Bombay <ul style="list-style-type: none"><li>One of five team members on a campus water-conservation initiative to retrofit tap aerators across institute buildings, reducing flow rate without affecting usability</li><li>Conducted <b>field surveys</b> to establish baseline consumption and liaised with <b>institute authorities</b> to secure approvals and coordinate installation</li></ul> |
| AUG 2015 – MAY 2016 | <b>A novel paper sensor as a diagnostic test for Multiple Sclerosis</b> <ul style="list-style-type: none"><li>Designed a <b>low-cost paper-based diagnostic</b> tool for multiple sclerosis, targeting accurate detection at a fraction of the cost of conventional assays</li><li>Presented at the Intel International Science and Engineering Fair 2016 in Phoenix, USA [<u>Category</u>: Translational Medical Science]</li></ul>                                    |

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| AUG 2014 –<br>MAY 2016 | <b>GGT inhibition – combating multi-drug resistance</b> <ul style="list-style-type: none"> <li>Developed a technique to inhibit <math>\gamma</math>-glutamyl transpeptidase, a candidate pathway in <b>bacterial multi-drug resistance</b>, as a route to restoring antibiotic sensitivity</li> <li>Presented at the Intel International Science and Engineering Fair 2015 in Pittsburgh, USA [Category: Microbiology]</li> <li>Invited by the Indian Science Congress (ISC'14) and won a special award presented by the Gujarat Council on Science and Technology (GUJCOST)</li> </ul> |
| AUG 2013 –<br>MAY 2014 | <b>Keratin waste – an effective management</b> <ul style="list-style-type: none"> <li>Converted <b>keratin waste</b> from leather industry into a <b>bio-fertilizer</b> through microbial degradation by <i>B. licheniformis</i> ER15, diverting a recalcitrant waste stream into agricultural use</li> <li>Presented at the Intel International Science and Engineering Fair 2014 in Los Angeles, USA [Category: Microbiology]</li> </ul>                                                                                                                                              |

## SKILLS

|             |                                                                                |
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| PROGRAMMING | Python, R, MATLAB, LaTeX                                                       |
| METHODS     | Machine learning, computational fluid dynamics, source apportionment (PMF/NMF) |
| SOFTWARE    | QGIS, ArcGIS, Origin                                                           |
| LANGUAGES   | English, Hindi                                                                 |

## ACADEMIC SERVICE & POSITIONS OF RESPONSIBILITY

|                        |                                                                                                                                                                                                                                                     |
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| MAY 2023 –<br>APR 2024 | <b>Department General Secretary</b> , ESED Council, Indian Institute of Technology Bombay<br>Coordinated the student council and convened technical seminars, cultural events, and industrial visits for the department consisting of 350+ students |
| AUG 2021 –<br>MAY 2022 | <b>Alumni Secretary</b> , ESED Council, Indian Institute of Technology Bombay<br>Coordinated alumni engagement through events and mentorship sessions                                                                                               |
| SEP 2019 –<br>JUN 2020 | <b>Internship Coordinator</b> , Institute Placement Cell, Indian Institute of Technology Bombay<br>One of 35 team members serving 1,900+ students; approached companies and universities to source internship opportunities for IIT Bombay students |

## ORGANIZATION OF CONFERENCES & PROFESSIONAL MEMBERSHIPS

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| 2026 –    | <b>Lifetime Member</b> , IIT Bombay Alumni Association, India                                |
| 2024 – 26 | <b>Organizer</b> , Annual Symposium on Advanced Air Quality Sensing, ANTL, IIT Bombay, India |
| 2024      | <b>Founder and Chief Organizer</b> , Tatva, ESED Annual Fest, IIT Bombay, India              |
| 2023 – 24 | <b>Student Member</b> , EGU and IASTA                                                        |

## REFERENCES

|                   |                                                                                                                                                                            |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PHD<br>SUPERVISOR | <b>Prof. Manoranjan Sahu</b><br>Associate Professor, Indian Institute of Technology Bombay, India<br><a href="mailto:mrsahu@iitb.ac.in">mrsahu@iitb.ac.in</a>              |
| RPC<br>MEMBER     | <b>Prof. Srinidhi Balasubramanian</b><br>Assistant Professor, Indian Institute of Technology Bombay, India<br><a href="mailto:srinidhi@iitb.ac.in">srinidhi@iitb.ac.in</a> |